

Long error bursts : $L-1 > n-k$

$$\begin{cases} e(x) = x^i (x^{L-1} + ??? + 1) \\ e(x) = c(x) \cdot g(x) \end{cases}$$

what is total # undetectable error bursts of L ? $\Rightarrow (n-L+1) \times 2^{(L-1)-(n-k)-1}$

$$c(x) \cdot g(x) = x^i (x^{L-1} + ??? + 1)$$

$$c(x) \cdot (\underbrace{x^{n-k} + \dots + 1}_{\text{given}}) = x^i (x^{L-1} + ??? + 1)$$

For $c(x)$:

① has x^i as its factor

$$\Rightarrow c(x) = x^i \cdot c'(x) = x^i (x^{(L-1)-(n-k)} + ??? + 1) \quad \begin{cases} i = 0, \dots, n-L \\ \text{"?" terms : } (L-1)-(n-k)-1 \end{cases}$$

② Highest degree term

$$i + \deg(c'(x)) + (n-k) = i + (L-1)$$

$$\Rightarrow \deg(c'(x)) = (L-1) - (n-k) \quad (*)$$

$$\downarrow \\ (n-L+1) \times 2^{(L-1)-(n-k)-1}$$

③ Lowest degree term

$$i + \text{lowest degree}(c'(x)) + 0 = i + 0$$

$$\Rightarrow \text{lowest degree}(c'(x)) = 0 \quad (*)$$

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FUE (L)

$$= \frac{\text{total \# undetectable } L}{\text{total \# error bursts of } L} = \frac{(n-L+1) \times 2^{(L-1)-(n-k)-1}}{(n-L+1) \times 2^{L-2}}$$

$$= 2^{-(n-k)} = \frac{1}{2^{n-k}}$$

(when $L-1 > n-k$)

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E.g. $g(x) = x^3 + x + 1$, $K=4$, $n=7$, $n-K=3$

$$\text{FUE}(L=6) = ? = \frac{1}{2^{n-K}} = \frac{1}{8} = \frac{(n-L+1) \cdot 2^{(L-1)-(n-K)-1}}{(n-L+1) \cdot 2^{L-2}} = \frac{2 \cdot 2^1}{2 \cdot 2^4} = \frac{4}{32}$$

$L-1 = 5 > 3 = n-K$ $\downarrow$ $\uparrow$

$L=6$: $e(x) = x^{\underline{i}} (x^{\underline{5}} + ??? + 1)$, $i=0,1$

$$\begin{aligned} g(x) \cdot c(x) &= (x^3 + x + 1) \cdot c(x) \\ &= (x^3 + x + 1) \cdot x^{\underline{i}} \cdot c'(x) \\ &= (x^3 + x + 1) \cdot x^{\underline{i}} \cdot (x^2 + ?x + 1) \end{aligned}$$

Pick $i=0, ?=0 \Rightarrow e(x) = (x^3 + x + 1) \cdot 1 \cdot (x^2 + 1)$
 $= x^5 + x^2 + x + 1 \Rightarrow e = [0100111]_{1x7}$

$i=0, ?=1 \Rightarrow e(x) = (x^3 + x + 1) \cdot x^0 \cdot (x^2 + x + 1)$
 $= x^5 + x^4 + 1 \Rightarrow e = [0110001]_{1x7}$

$i=1, ?=0 \Rightarrow e = [1001110]_{1x7}$

$i=1, ?=1 \Rightarrow e = [1100010]_{1x7}$

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* $(L-1) < (n-K)$ "short"
 $\text{FUE}(L) = 0$

* $(L-1) > (n-K)$ "long"
 $\text{FUE}(L) = 1/2^{n-K}$

* $(L-1) = n-K$ "special" $\text{FUE}(L) = \frac{(n-K+1)}{(n-K+1) \cdot 2^{L-2}} = \frac{1}{2^{L-2}}$

$$\begin{cases} e(x) = x^{\underline{i}} \cdot (x^{\underline{L-1}} + ??? + 1) = x^{\underline{i}} \cdot (x^{n-K} + ??? + 1) \\ g(x) = x^{n-K} + \dots + 1 \end{cases} = \frac{1}{2^{(n-K-1)}}$$

$\Rightarrow e(x) = g(x) \cdot c(x)$

$$\begin{aligned} x^{\underline{i}} \cdot (x^{n-K} + ??? + 1) &= (x^{n-K} + \dots + 1) \cdot c(x) \\ &\Downarrow \\ x^{\underline{i}} \cdot c'(x) & \\ &\Downarrow \\ c'(x) &= 1 \\ &\Downarrow \\ c(x) &= x^{\underline{i}} \end{aligned}$$

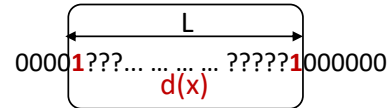
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Error Detection Capabilities

⊕ For Error Bursts of Length L:

➡ For error burst starting at bit location i and ending at bit location $(i + L - 1)$

- $e(x) = x^{i+L-1} + \dots + x^i = x^i d(x)$ where $d(x) = x^{L-1} + \dots + 1$



➡ $g(x)$ has degree $(n - k)$

- $L < (n - k + 1)$ *short*
 - $g(x)$ cannot divide $d(x)$ because $\deg(d(x)) < \deg(g(x))$
 - Can detect all such error bursts
- $L = (n - k + 1)$ *special*
 - $d(x)$ is divisible by $g(x)$ if and only if $d(x) = g(x)$
 - Fraction of such error bursts that are undetectable is $\frac{1}{2}^{(n-k-1)}$
- $L > (n - k + 1)$ *long*
 - Fraction of such error bursts that are undetectable is $\frac{1}{2}^{(n-k)}$

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Overall FUE for CRC code :

$\begin{cases} k \text{ info bits} \Leftarrow \\ (n-k) \text{ check bits} \\ n \text{ codeword bits} \end{cases}$

$$B = \left[\underbrace{\quad\quad\quad}_k \quad \underbrace{\quad\quad\quad}_{n-k} \right]_{1 \times n}$$

2^k : # diff codewords

$$\text{Overall FUE} = \frac{\text{total \# undetectable errors}}{\text{total \# valid errors}} = \frac{2^k - 1}{2^n - 1} < \frac{2^k - 1}{2^n - 2^{n-k}} = \frac{2^k}{2^{n-k}(2^k - 1)} = \frac{1}{2^{n-k}}$$

$$R = B + e$$

e is undetectable

$\Rightarrow$ Pattern in "R" $\Rightarrow$ "R" is codeword but different from "B"

$\Rightarrow 2^k - 1$ such cases

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